

Tumor Detection using AI

By : Aryan Jadhav
Soham Bhosage
Ganraj Kendre
Varnan Dhondage



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NEED OF PROJECT



- AI can “see” things that we humans can’t, and can find complex patterns and relationships between very different kinds of data.
- The accuracy of AI is reliable enough to be implemented in real life.
- AI can be more reliable when it comes to identifying if a tumor is cancerous or not.
- This can help radiologists with the identification of tumors.
- This can also help the doctors, as AI can help to identify some specific features or symptoms of the tumor.
- This can further help doctors to learn about Cancer.

INTRODUCTION



- Cancer is the leading cause of death in the world today
- We have proposed a methodology to detect cancer using AI
- This system will help the radiologists to detect cancer more reliably
- In this system we analyse the data input by user and the system will give us a conclusion if the given tumor is malicious or benign.
- The data given by user is in the form of image or in the form of textual reports.

PROBLEM DEFINITION



- Cancer is the leading cause of death in the world.
- This disease is characterized by the development of abnormal cells that divide uncontrollably and the ability to infiltrate and destroy normal body tissue
- Some general signs and symptoms associated with, but not specific to, cancer, include:
 - Fatigue.
 - Lump or area of thickening that can be felt under the skin.
 - Weight changes, including unintended loss or gain.
 - Changes in bowel or bladder habits
- have

PROPOSED METHODOLOGY

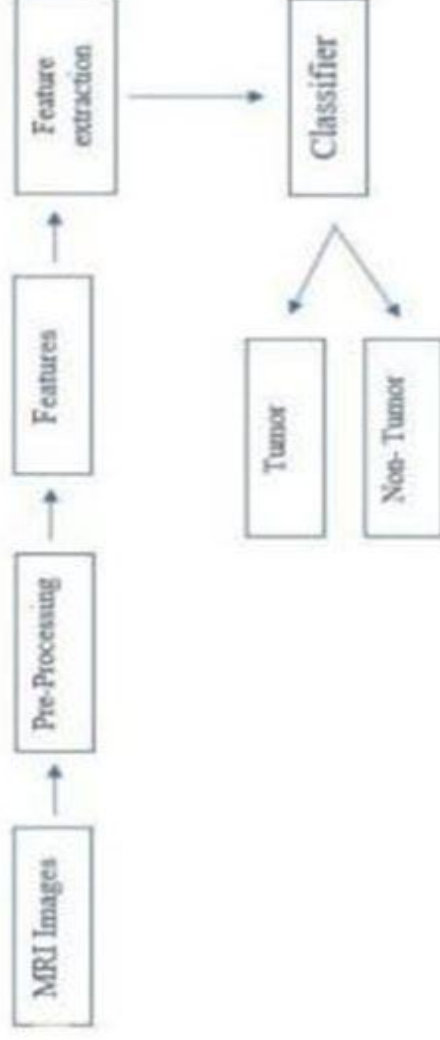


The technique used for making this model work is Neural Network The different stages by which a Neural Network is applied to the dataset are:

- Import the required packages
- Import the data sets
- Image reading is done after that it is labeled (such as 0 for benign and 1 for malicious) and then the reports are stored in the data frame.
- The training sets are designed in a way that the AI algorithm learns from the training sets
- The Accuracy of the System is evaluated
- The loss graph and the accuracy graph are plotted.



System Architecture

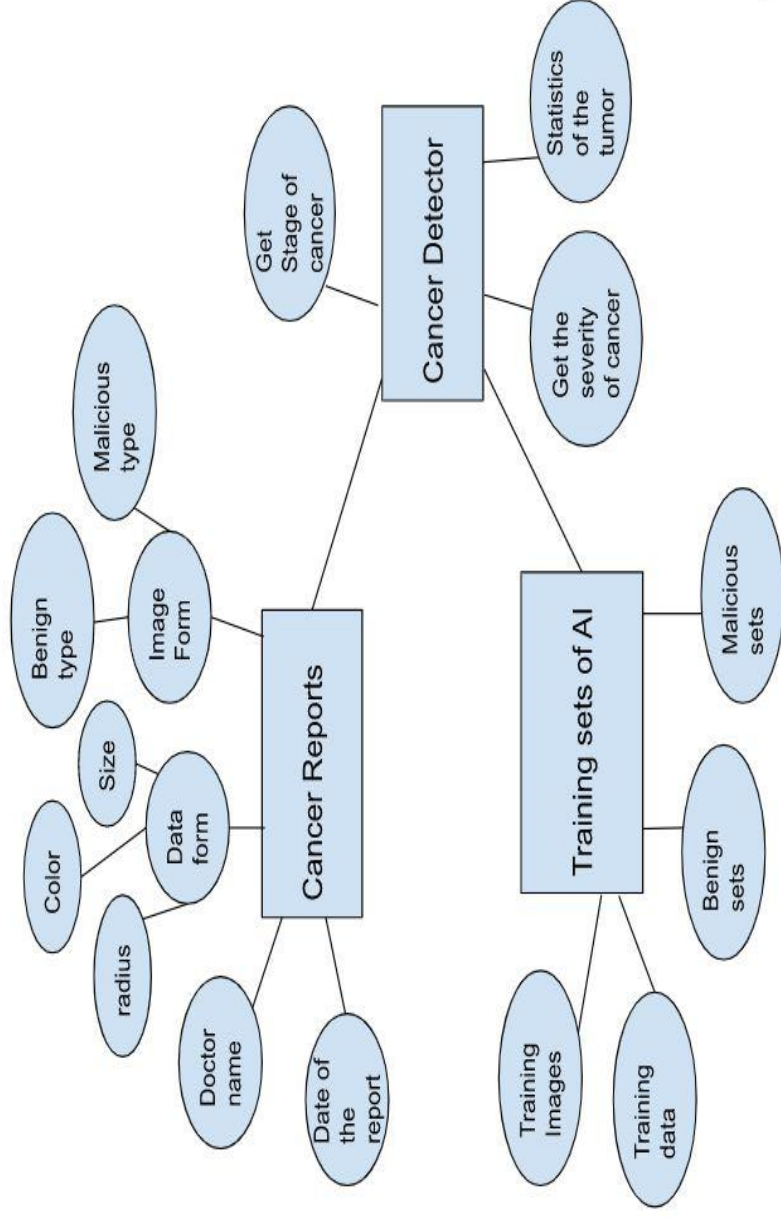


RESOURCES REQUIRED

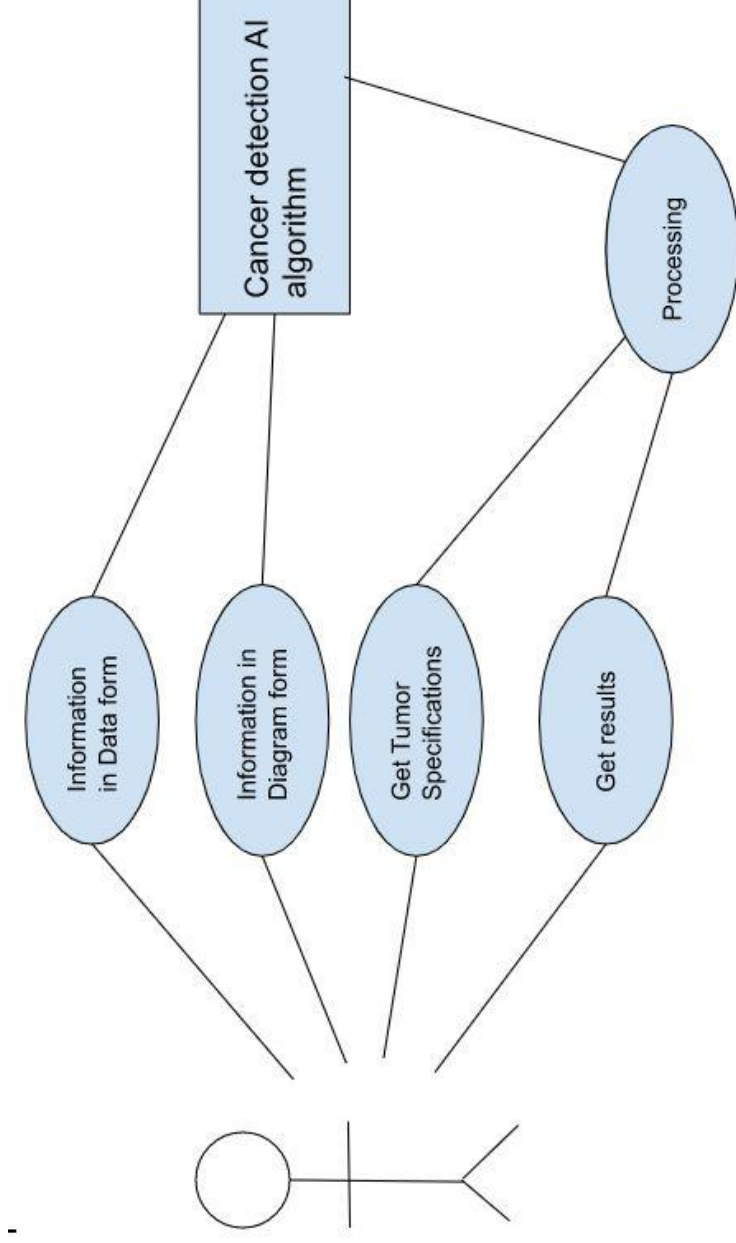


RAM	8GB
HARD DISK	40GB
KEY BOARD	Standard windows keyboard
MOUSE	Two or three button mouse
MONITOR	LCD/LED
OPERATING SYSTEM	WINDOWS 11
CODING LANGUAGE	PYTHON
EDITOR	PYCHARM OR VS CODE

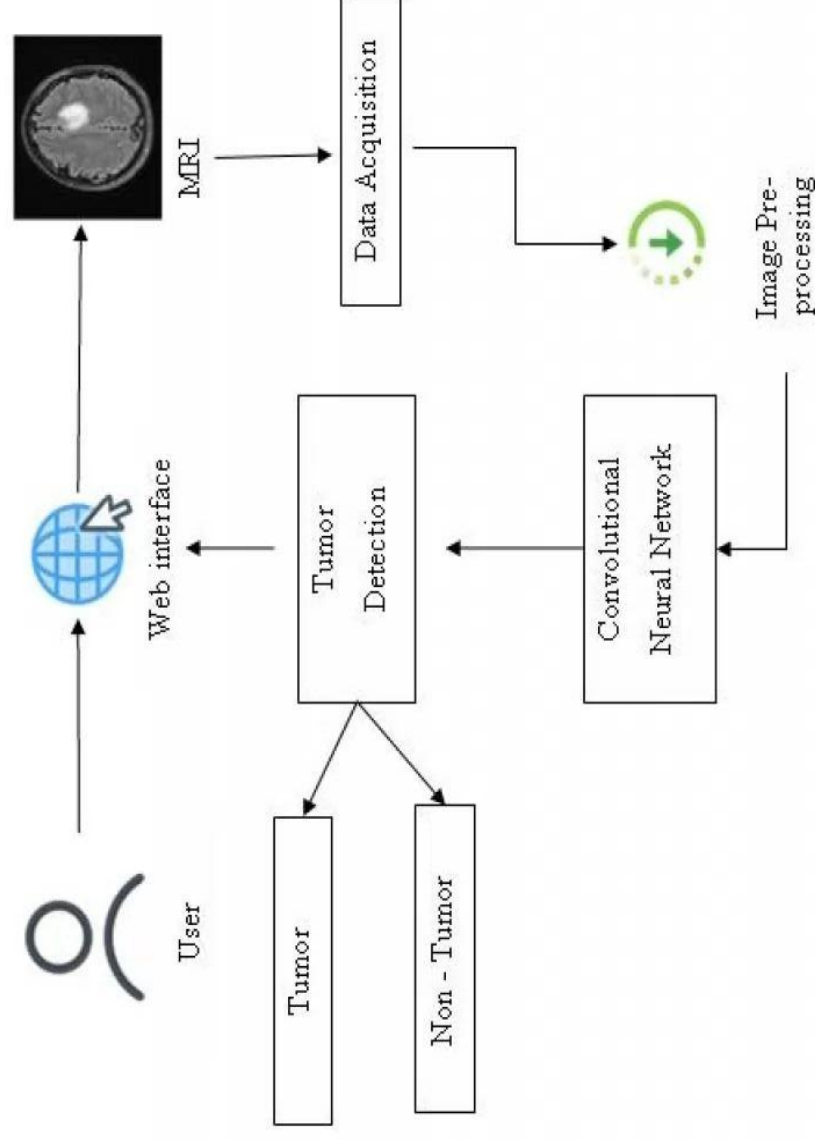
ER DIAGRAM



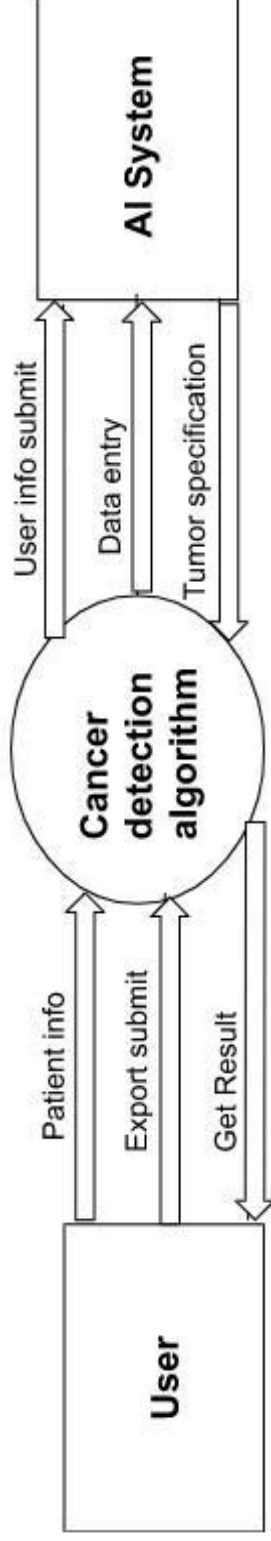
USE CASE DIAGRAM



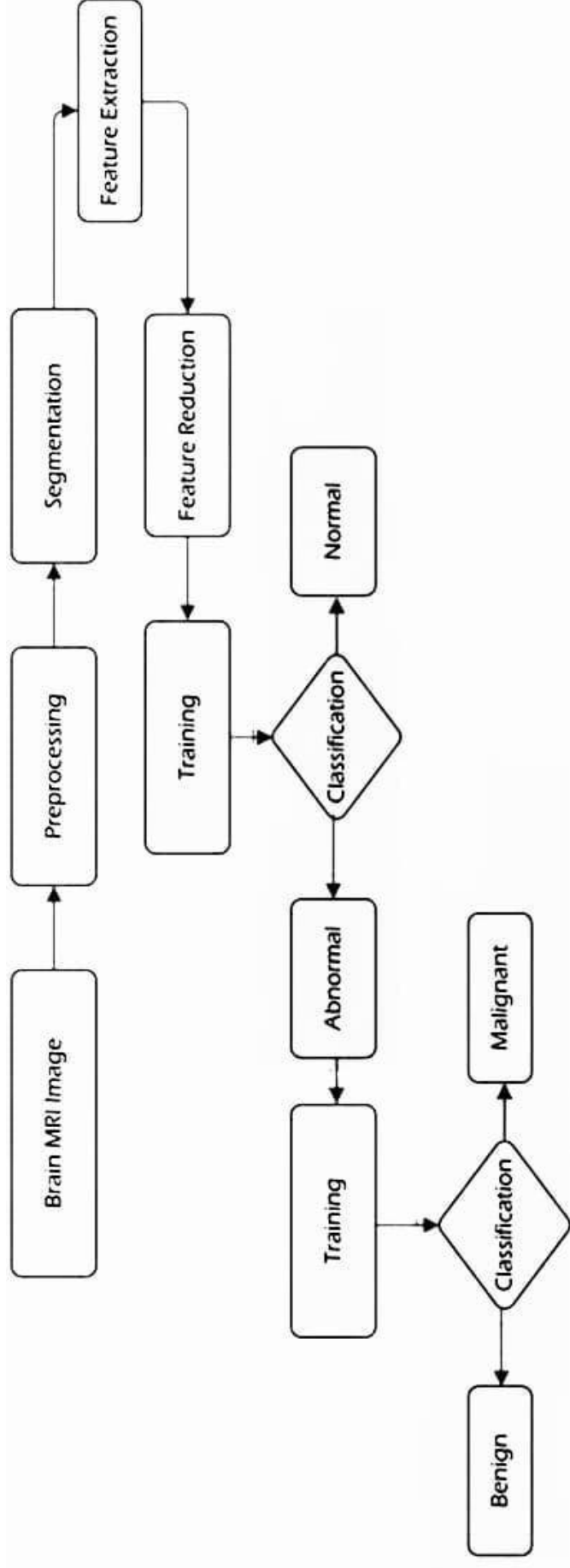
ACTIVITY DIAGRAM



DFD DIAGRAM: LEVEL 0



DFD DIAGRAM: Level 1



Thank
you